On-boarding and Off-boarding simulation locally

The domain I created is LAB.local. I have the default Domain Controllers OU right now. The OU structure for this LAB simulation will look like this:

LAB.local

- Domain Controllers
- LAB users
    - New Hires
    - Disabled Users
- LAB Groups

Creating the parent OU, the child Ous under LAB Users and the groups OU using PoIrShell commands:

```

# Create parent OU

New-ADOrganizationalUnit -Name "LAB Users" -Path "DC=LAB,DC=local"


# Create child OUs under LAB Users

New-ADOrganizationalUnit -Name "New Hires" -Path "OU=LAB Users,DC=LAB,DC=local"

New-ADOrganizationalUnit -Name "Disabled Users" -Path "OU=LAB Users,DC=LAB,DC=local"


# Create groups OU

New-ADOrganizationalUnit -Name "LAB Groups" -Path "DC=LAB,DC=local"
```

Verification:

```
PS C:\WINDOWS\system32> Get-ADOrganizationalUnit -Filter * | Select-Object Name

Name
----
Domain Controllers
LAB Users
New Hires
Disabled Users
LAB Groups
```

Next I will create security groups inside LAB groups (IT,HR, Finance) and the CSV templates that simulate the HR data feed:

```

# Create department groups inside LAB Groups OU

New-ADGroup -Name "IT" -GroupScope Global -GroupCategory Security -Path "OU=LAB Groups,DC=LAB,DC=local"

New-ADGroup -Name "HR" -GroupScope Global -GroupCategory Security -Path "OU=LAB Groups,DC=LAB,DC=local"

New-ADGroup -Name "Finance" -GroupScope Global -GroupCategory Security -Path "OU=LAB Groups,DC=LAB,DC=local"


# Verify

Get-ADGroup -Filter * -SearchBase "OU=LAB Groups,DC=LAB,DC=local" | Select-Object Name, GroupScope

```

```
PS C:\WINDOWS\system32> Get-ADGroup -Filter * -SearchBase "OU=LAB Groups,DC=LAB,DC=local" | Select-Object Name, GroupScope

Name    GroupScope
----    ----------
IT          Global
HR          Global
Finance     Global
```

Building the CSV now which is the HR data feed the script will read from

```
PS C:\WINDOWS\system32> New-Item -ItemType Directory -Path "C:\Scripts\Onboarding"


    Directory: C:\Scripts


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----          4/4/2026   2:00 PM                Onboarding
```

Where the project files will live.

```
PS C:\WINDOWS\system32> New-Item -Path "C:\Scripts\Onboarding\new_hires.csv" -ItemType "File"


    Directory: C:\Scripts\Onboarding


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----          4/4/2026   2:03 PM              0 new_hires.csv
```

This csv file is where the new hires data will live:

```
PS C:\WINDOWS\system32> Import-Csv "C:\Scripts\Onboarding\new_hires.csv"


FirstName  : John
LastName   : Smith
Department : IT
JobTitle   : Helpdesk Technician
Manager    : Administrator

FirstName  : Jane
LastName   : Doe
Department : HR
JobTitle   : HR Coordinator
Manager    : Administrator

FirstName  : Mike
LastName   : Johnson
Department : Finance
JobTitle   : Accountant
Manager    : Administrator

FirstName  : Sarah
LastName   : Williams
Department : IT
JobTitle   : Systems Administrator
Manager    : Administrator

FirstName  : Lisa
LastName   : Brown
Department : HR
JobTitle   : Recruiter
Manager    : Administrator
```

*Note: Manager is set to Administrator for all because it's the only user in this lab environment.

Time to build the Onboarding Script that will read new_hires.csv and create AD accounts:

Script is separate -> **Onboarding.ps1**

Output:

```
PS C:\Scripts\Onboarding> cat .\onboarding_log.txt
2026-04-05 21:23:19 - Starting onboarding run. 5 user(s) found in CSV.
2026-04-05 21:25:20 - Starting onboarding run. 5 user(s) found in CSV.
2026-04-05 21:25:20 - CREATED - jsmith (John Smith)
2026-04-05 21:25:20 - Group - jsmith added to IT group.
2026-04-05 21:25:20 - CREATED - jdoe (Jane Doe)
2026-04-05 21:25:20 - Group - jdoe added to HR group.
2026-04-05 21:25:20 - CREATED - mjohnson (Mike Johnson)
2026-04-05 21:25:20 - Group - mjohnson added to Finance group.
2026-04-05 21:25:20 - CREATED - swilliams (Sarah Williams)
2026-04-05 21:25:20 - Group - swilliams added to IT group.
2026-04-05 21:25:21 - CREATED - lbrown (Lisa Brown)
2026-04-05 21:25:21 - Group - lbrown added to HR group.
2026-04-05 21:25:21 - Onboarding run complete.
```

Let's verify the accounts are under New Hires OU:

```
PS C:\WINDOWS\system32> Get-ADUser -Filter * -SearchBase "OU=New Hires,OU=LAB Users,DC=LAB,DC=local" -Properties Department | Select-Object Name, SamAccountName, Department

Name           SamAccountName Department
----           -------------- ----------
John Smith     jsmith         IT
Jane Doe       jdoe           HR
Mike Johnson   mjohnson       Finance
Sarah Williams swilliams      IT
Lisa Brown     lbrown         HR
```

Testing duplicates:

```
PS C:\Scripts\Onboarding> .\Onboarding.ps1
2026-04-05 21:41:31 - Starting onboarding run. 5 user(s) found in CSV.
2026-04-05 21:41:31 - Skipped - jsmith already exists in AD.
2026-04-05 21:41:31 - Skipped - jdoe already exists in AD.
2026-04-05 21:41:31 - Skipped - mjohnson already exists in AD.
2026-04-05 21:41:31 - Skipped - swilliams already exists in AD.
2026-04-05 21:41:31 - Skipped - lbrown already exists in AD.
2026-04-05 21:41:31 - Onboarding run complete.
```

So far, I read a CSV simulated an HR data feed, auto-generated usernames/passwords (the user is req to change on login), skips duplicates safely, creates fully attributed AD accounts, assigns department group membership and logs every action with timestamps. Now it's time to simulate offboarding an employee or two.

I added one more employee, which is myself as an Intern for the Finance department. I will terminate myself as a part of the offboarding process. (my internship has come to an end)

```
PS C:\Scripts\Onboarding> Get-ADUser -Filter * -SearchBase "OU=New Hires,OU=LAB Users,DC=LAB,DC=Local" -Properties Department, Title | Select Name, SamAccountName, Department, Title

Name            SamAccountName Department Title
----            -------------- ---------- -----
John Smith      jsmith         IT         Helpdesk Technician
Jane Doe        jdoe           HR         HR Coordinator
Mike Johnson    mjohnson       Finance    Accountant
Sarah Williams  swilliams      IT         Systems Administrator
Lisa Brown      lbrown         HR         Recruiter
Gabriel Marrero gmarrero       Finance    Intern
```

I created an Offboarding folder where terminated_users.csv will live for the offboarding process:

```
PS C:\Scripts\Offboarding> Import-Csv ".\terminated_users.csv"

SamAccountName
--------------
jsmith
gmarrero
```

Script lives at **Offboarding.ps1**. Here is the output:

```
PS C:\Scripts\Offboarding> .\Offboarding.ps1
2026-04-06 13:14:54 - Starting offboarding run. 2 user(s) found in CSV.
2026-04-06 13:14:54 - Disabled - jsmith
2026-04-06 13:14:55 - Group Removed - jsmith removed from CN=IT,OU=LAB Groups,DC=LAB,DC=local.
2026-04-06 13:14:55 - Moved - jsmith moved to Disabled Users OU.
2026-04-06 13:14:55 - Disabled - gmarrero
2026-04-06 13:14:55 - Group Removed - gmarrero removed from CN=Finance,OU=LAB Groups,DC=LAB,DC=local.
2026-04-06 13:14:55 - Moved - gmarrero moved to Disabled Users OU.
2026-04-06 13:14:55 - Offboarding run complete.
```

Let's confirm both users are disabled, no longer in their groups, and moved to the correct Disabled Users OU:

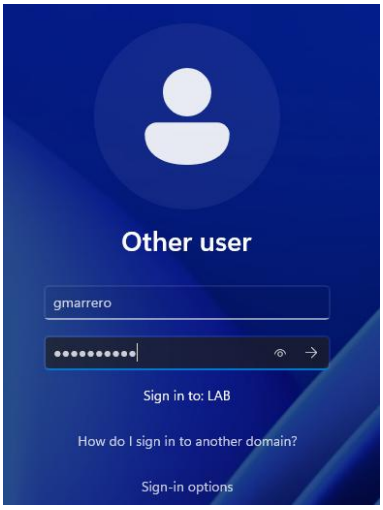
```
PS C:\Scripts\Offboarding> Get-ADUser -Filter * -SearchBase "OU=Disabled Users,OU=LAB Users,DC=LAB,DC=local" -Properties Enabled | Select-Object Name, SamAccountName, Enabled

Name            SamAccountName Enabled
----            -------------- -------
John Smith      jsmith           False
Gabriel Marrero gmarrero         False
```

Checking the IT group and Finance group respectively to make sure both users don't exist:

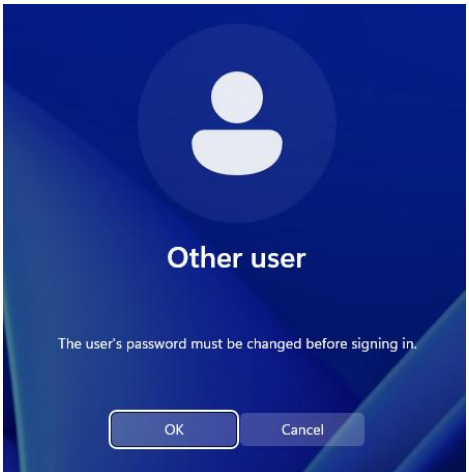
```
PS C:\Scripts\Offboarding> Get-ADGroupMember -Identity "IT" | Select-Object Name, SamAccountName

Name           SamAccountName
----           --------------
Sarah Williams swilliams


PS C:\Scripts\Offboarding> Get-ADGroupMember -Identity "Finance" | Select-Object Name, SamAccountName

Name         SamAccountName
----         --------------
Mike Johnson mjohnson
```

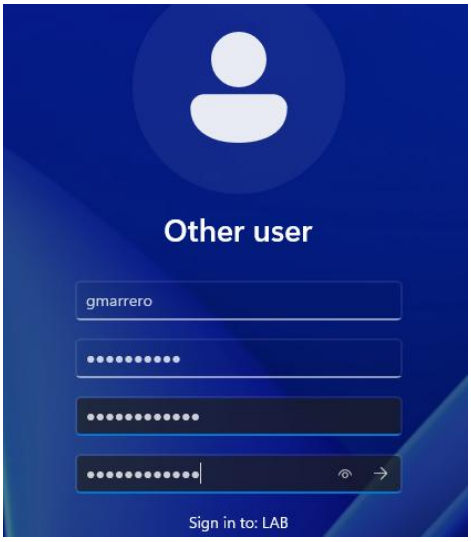
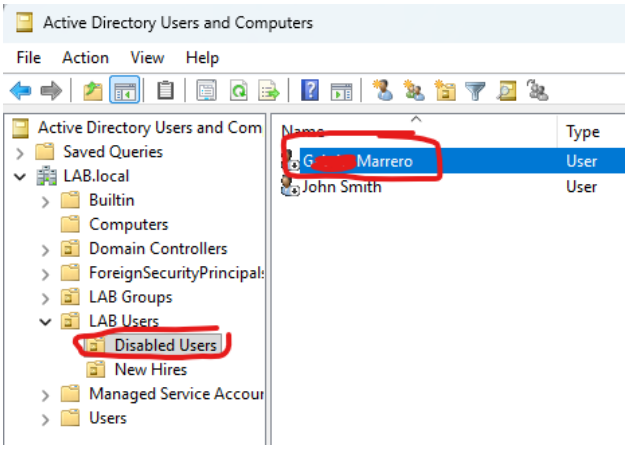
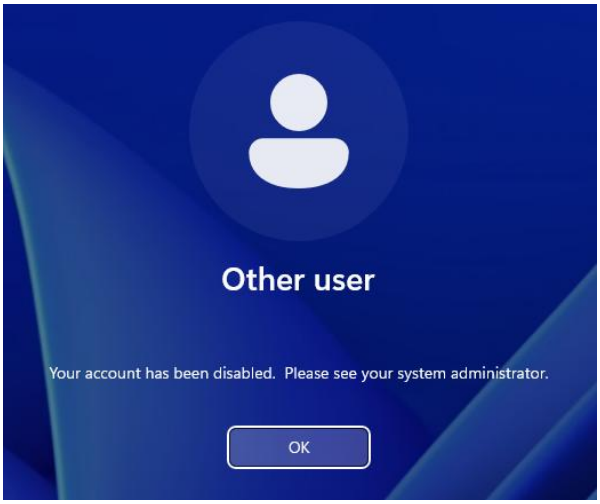
I will be testing the Intern employee, gmarrero

- I should be able to change password at logon
- When I offboard gmarrero, the account should show clearly that it is disabled, and I can no longer login



Works as intended:





Password has changed successfully and I can login. Now let's test the offboard/disable operation: